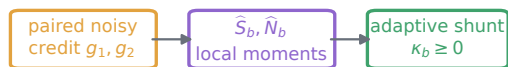


A

Local reliability estimate



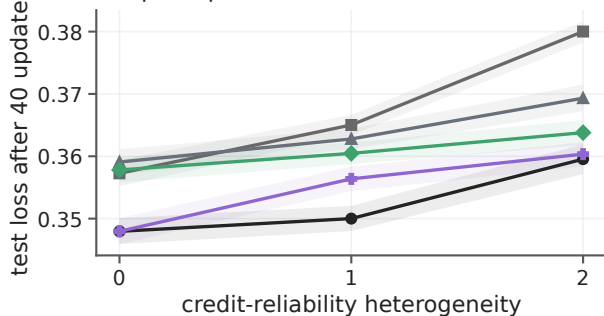
$$\hat{S}_b = (g_1, g_2) \quad \hat{N}_b = \|g_1 - g_2\|^2/2$$

$$a_b = \min\{1, \hat{S}_b/[c(\hat{S}_b + \hat{N}_b)]\}$$

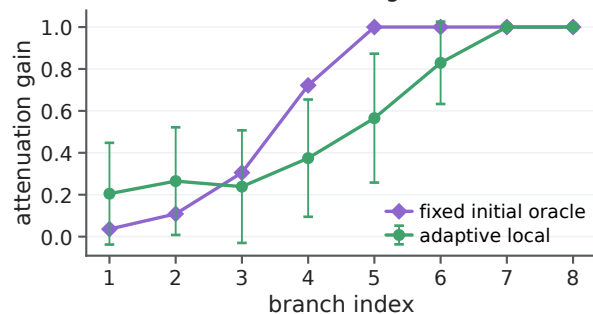
C

test loss after 40 updates

Adaptive placement is conditional

**B**

Estimated branch ordering

**D**

control loss - adaptive-local loss

Final-loss boundary at high heterogeneity

